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# AI Veterinary Assistance: Enhancing Clinical Decision-making in Animal Healthcare

**Youn-Gyu Jin [1,†], Guixin Wu [1,†], Ju-Won Seo [1], Seong-Jin Park [1], Sung-Ho Hur [2], Dinara Aliyeva [3], Jun-Hyung Park [4], and Kang-Min Kim [1,5]**

[1]Department of Artificial Intelligence, The Catholic University of Korea, Bucheon-si 14662, Republic of Korea
[2]IntoCNS Inc., Room 2101, Building A, 767, Sinsuro, Suji-gu, Yongin-Si, Gyeonggi-do, Republic of Korea
[3]Department of Computer Science, College of Arts & Sciences, The University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA
[4]Division of Language & AI, Hankuk University of Foreign Studies, Seoul 02450, Republic of Korea
[5]Department of Data Science, The Catholic University of Korea, Bucheon-si 14662, Republic of Korea
[†]These authors contributed equally to this work.

Corresponding author: Kang-Min Kim (kangmin89@catholic.ac.kr).

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**ABSTRACT** As the number of households raising companion animals increases, there is a corresponding increase in the demand for veterinary medical services. Furthermore, as companion animals age, the prevalence of chronic disease is increasing, leading companion animal owners to seek more help with long-term health management. However, the shortage of veterinarians and the growing complexity of the consultation process have led to a heavier workload for veterinarians, potentially hindering clinical decision-making efficiency. To address these challenges, we propose **AI V**eterinary **A**ssistance (AVA), a framework leveraging a large language model. AVA automatically extracts symptoms from consultation records, predicts most probable disease using veterinarian certified disease-symptom database and recommend future consultation question. AVA achieved disease prediction accuracies of 91.4%, 93.4%, and 95.9% for Top-3, Top-5, and Top-10, respectively, along with a symptom extraction accuracy of 79.9% on a consultation records dataset constructed from a veterinarian-certified disease-symptom database. Furthermore, on a real-world dataset, AVA achieved disease prediction accuracies of 38.6%, 43.6%, and 51.5% for Top-3, Top-5, and Top-10, respectively. In both datasets, AVA outperformed baseline methods, demonstrating its effectiveness in supporting clinical decision-making. These results suggest that AVA can help reduce veterinarians' workload while enhancing the efficiency and reliability of the consultation process.

**INDEX TERMS** Clinical Decision-making Assistance, In-Context Learning, Large Language Model

## I. INTRODUCTION

As living standards improve and awareness of companion animals increases, the number of households with companion animals is rising globally [1], [2], [3]. The COVID-19 pandemic further accelerated this trend, as prolonged lockdowns led to a significant increase in companion animal adoptions [4], [5], [6], [7]. In 2024, approximately 66% of households in the United States had a companion animal [8], while around 50% of European households did so in 2022 [9]. Similarly, in 2022, an average of 59% of households across 12 Asian countries, including China, Japan, and Vietnam, had a companion animal [10].

Among the various factors, household structure significantly influences companion animal ownership decisions, with married couples without children exhibiting the highest likelihood of companion animal adoption [1]. South Korea, which currently has the lowest birth rate in the world [11], has seen many married couples choosing companion animal adoption as an alternative to having children. Although South Korea's companion animal ownership rate (28.2% as of 2022) remains lower than the Asian average, it has been rapidly increasing with a remarkable 62% growth over the past 13 years [12]. However, infrastructure development has not kept pace with this rapid growth, leading to various challenges related to overcrowding, particularly as most companion animal-owning households are concentrated in the Seoul metropolitan area [13].

The rising popularity of companion animal ownership has also increased demand for veterinary medical services. Many countries have reported a rise in the frequency of veterinary service utilization, along with an increase in companion animal insurance subscriptions which has contributed to the

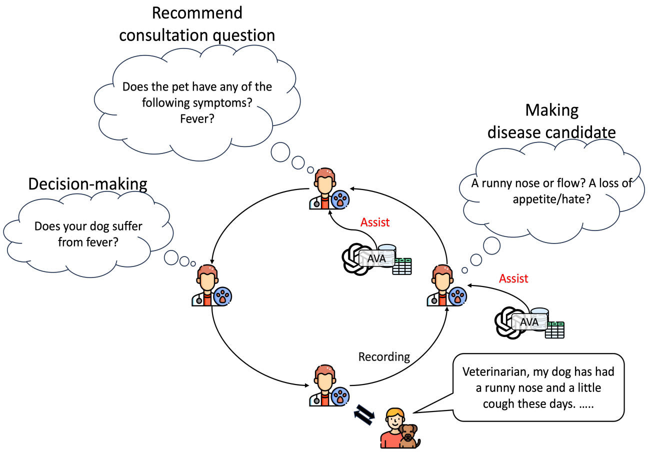


**FIGURE 1. Overview of manual and AVA assisted veterinary consultation processes.**

growth of the veterinary services market [14], [15], [16]. The global veterinary services market is projected to grow steadily at a compound annual growth rate of 5.92% from 2025 to 2034 [17]. A similar trend is observed in South Korea, where dog owners visit veterinary clinics an average of 4.47 times per year [18]. In addition, companion animal healthcare expenditures have shown steady growth, reaching an average of approximately $435 per companion animal in 2023 [12].

As a result, veterinarians are facing an increasing number of cases. This growing workload has resulted in shorter average consultation times per companion animal, which negatively impacts both diagnostic accuracy and companion animal owner satisfaction [19], [20]. During consultations, veterinarians rely on companion animal owners' descriptions of their companion animals' conditions to identify key symptoms, predict possible diseases, and determine the need for additional examinations or inquiries (Figure 1). Although the final diagnosis remains the veterinarian's responsibility, extracting key symptom information from conversations with companion animal owners and leveraging veterinary data to suggest potential diseases can be supported through an artificial intelligence (AI)-based system. Such a support system can significantly alleviate veterinarians' workload.

Various systems have been developed to assist veterinarians, but most have focused on specific medical procedures or the diagnosis of individual diseases [21], [22]. There is a lack of systems designed to support veterinarians during the early stages of consultation, where they must infer numerous potential diseases and symptoms based on owners' descriptions. In addition, for a diagnostic support system to gain veterinarians' trust, it must provide transparency in its reasoning and decision-making process. Explainable AI (XAI) is essential in this context, yet the development of veterinary diagnostic support systems incorporating XAI remains limited [23].

In contrast, human medicine has seen significant advancements in consultation assistance systems using natural language processing (NLP) techniques. Ongoing research has explored the application of electronic health record (EHR) data to support medical consultations [24], [25], [26]. Recent developments in large language models (LLMs) [27], such as ChatGPT, have accelerated the creation of AI technologies capable of analyzing medical data and assisting in consultations [28], [29]. One of the main challenges in applying these technologies to veterinary medicine is that, unlike large medical institutions in human healthcare, most veterinary clinics operate on a smaller scale and typically lack sufficient computing infrastructure such as dedicated servers or GPUs. These limitations, caused by space and cost constraints, make computational efficiency a critical concern in implementing deep learning-based systems. This applies not only to training models from scratch but also to running LLMs, as inference alone may exceed the computing capacity available in typical veterinary settings.

To address these challenges, we propose a novel framework for real-time analysis of veterinary consultation records, termed the **A**I **V**eterinary **A**ssistance (AVA). Built upon the latest LLM, GPT-4o [30], AVA aims to enhance consultation assistance by integrating a veterinary disease-symptom database, constructed from Korean companion animal EHR data, with an LLM. The framework operates in two stages. In the first stage, GPT-4o automatically extracts symptom-related information from veterinary consultation records. In the second stage, the extracted symptoms are matched with the disease-symptom database to predict possible diseases and recommend additional consultation questions for further inquiry. This approach not only improves the effectiveness of real-time consultations but also provides veterinarians with clear explanations of the system's predictions, ensuring transparency in clinical decision-making.

Furthermore, AVA is designed for seamless integration into general veterinary clinics by accommodating variations in terminology and documentation formats. Its database-driven architecture allows for easy updates to incorporate new diseases and symptoms, ensuring long-term scalability while minimizing maintenance efforts. Although the disease-symptom database and the data used for development were constructed from Korean veterinary EHR data, the technology itself is not language-dependent. As long as an appropriate database is established, AVA can be widely applied in various linguistic and regional contexts.

To evaluate AVA's effectiveness, we conducted experiments using consultation records generated from a veterinarian-certified disease–symptom database, as well as real-world consultation data. On the generated consultation records, AVA achieved a symptom extraction accuracy of 79.9% and disease prediction accuracies of 91.4%, 93.4%, and 95.9% for Top-3, Top-5, and Top-10, respectively. In the evaluation using real-world consultation records, AVA consistently outperformed baseline methods. Without any additional fine-tuning, AVA demonstrated stable performance in Top-$K$ prediction tasks, supporting its potential as an effective clinical decision-support system.

In summary, our contributions are three-fold:

- We propose AVA, an LLM-based framework designed to assist decision-making in veterinary consultations by analyzing consultation records, predicting diseases, and recommending consultation questions to enhance consultation efficiency.
- We construct a veterinary-specific disease-symptom database that enables AVA to support more precise and explainable decision-making, thereby effectively assisting veterinarians during consultations.
- We demonstrate the effectiveness of AVA by showing that it outperforms baselines in both consultation records and real-world datasets.

## II. RELATED WORK

### A. LARGE LANGUAGE MODEL

LLMs have driven significant advancements in NLP technology. From general text generation to contextual understanding and complex logical reasoning, the latest LLMs exhibit human-like language processing capabilities, making them valuable tools across various fields [31], [32], [33], [34]. These models internalize extensive knowledge by training on vast datasets, enabling them to adapt to new tasks with minimal data and even infer concepts they have not been explicitly trained on, due to their few-shot and zero-shot learning capabilities [35], [36]. However, general-purpose LLMs are predominantly trained on publicly available data, such as web documents, news articles, and encyclopedic content. This limits their effectiveness in specialized domains such as medicine, law, and veterinary science [37]. To address this limitation, domain adaptation is a promising approach, though not without challenges. The process must account for risks such as dataset bias, overfitting to narrow distributions, and inconsistencies in terminology or documentation styles. Nevertheless, when performed with expert-verified data, domain adaptation can substantially enhance language understanding and reasoning [38], [39]. Several LLMs specialized for the medical domain [40], [41], [42], [43] demonstrate superior language comprehension and reasoning capabilities, achieving strong performance in medical tasks. Notably, these domain-specific LLMs outperform general models in executing specialized tasks more efficiently, making them particularly valuable for real-time query response and decision-making assistance [44], [45]. This characteristic highlights their potential for real-time applications in veterinary medicine, such as consultation assistance and symptom-based disease prediction. However, the lack of domain expertise in veterinary decision-making, the black-box nature of LLMs, and the risk of hallucinations pose challenges to their application in the veterinary domain. In particular, hallucinations in clinical settings can lead to unsafe treatment suggestions or misinterpretation of symptoms, potentially compromising patient safety [46]. In addition, the opaque reasoning behind LLM outputs, often described as their black box nature, undermines trust and accountability in high stakes domains like veterinary medicine. To address these limitations, this study integrates an LLM with a veterinary disease-symptom database, aiming to supplement insufficient domain knowledge and provide more reliable and transparent assistance in clinical decision-making.

### B. INFORMATION EXTRACTION USING LANGUAGE MODEL

Information extraction (IE) is a key technology for automatically extracting meaningful information from large volumes of text data. It has been actively researched across various domains, including medicine, law, finance, and science [47], [48], [49], [50]. Early IE systems primarily relied on rule-based approaches that extracted information using predefined patterns and rules [51], [52], [53]. However, these methods faced limitations in scalability and adaptability to new data environments [54]. With advancements in LLMs, research on LLM-based IE has expanded rapidly [55]. Recent studies have explored models trained on domain-specific data to improve extraction accuracy [55], while others have leveraged LLM-based approaches for IE across various fields, including medicine [56]. Notably, LLMs have been applied to extract medical information such as symptoms, diagnoses, and medications from unstructured texts, including EHRs [57], [58]. These findings suggest that, despite being trained on general-purpose datasets, LLMs can effectively analyze medical data and perform IE tasks. In-context learning (ICL) has emerged as a crucial technique for conducting IE with LLMs [59]. ICL enables models to perform specific tasks by providing only a small number of examples, eliminating the need for additional fine-tuning, and it has shown strong performance across various NLP applications in medicine, law, finance, and science [60], [61], [62], [63]. However, LLM-based ICL does not always guarantee high performance, especially in cases involving ambiguous expressions, out-of-distribution inputs, or concepts that are underrepresented in the training data. These limitations can hinder the accuracy and reliability of information extraction across various domains, including veterinary medicine. To address these limitations, this study proposes an approach that leverages customized prompts grounded in a veterinary disease symptom database. By incorporating rare symptoms and ambiguous expressions into the database in advance, our method enables more precise and dependable symptom extraction and disease prediction. This approach mitigates the generalization limitations of LLMs and supports the development of a practical consultation assistance system tailored to the veterinary domain.

| Diseases \ Symptoms | Vomiting | Bloody vomiting | Anuria | ... |
|---|---|---|---|---|
| Gastrinoma | 1 | 1 | 0 | ... |
| Corneal ulcer | 0 | 0 | 0 | ... |
| Keratitis | 0 | 0 | 0 | ... |
| Portosystemic shunt | 1 | 0 | 0 | ... |
| Liver failure acute | 1 | 0 | 0 | ... |
| Hepatitis acute | 1 | 0 | 0 | ... |
| Hepatitis chronic | 1 | 0 | 0 | ... |
| Common cold | 0 | 0 | 0 | ... |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |

**FIGURE 2. Partial display of the database. The database is stored in the form of a table, with the header as symptoms and the index column as diseases. If the number in the cell is 1, it indicates that the corresponding disease and symptom are related, and if it is 0, it indicates that there is no relationship.**

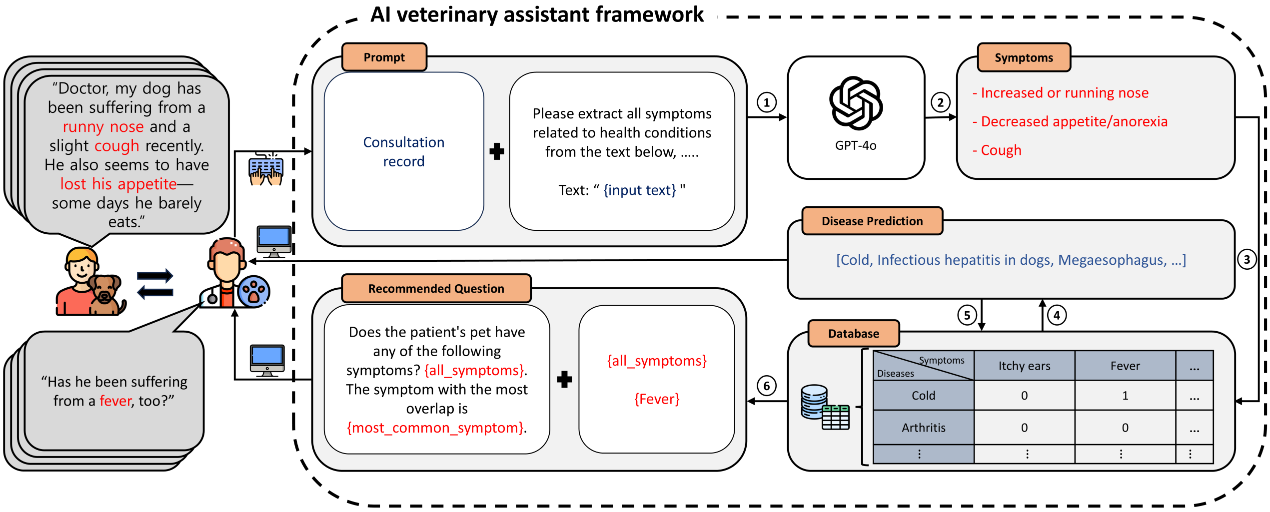


**FIGURE 3. Overview of AVA framework. Our framework utilizes LLM to extract symptoms from consultation records. It then performs disease prediction and recommends consultation questions by accessing a reliable and scalable disease-symptom database.**

**TABLE 1. The prompt used to construct the generated dataset.**

| | Prompt |
|---|---|
| **Data Generation** | You are a veterinarian. Based on the given symptoms, generate a natural and coherent consultation record.<br>**Important Guidelines:**<br>• Include all selected symptoms.<br>• Avoid simple enumeration and use varied expressions for each symptom.<br>• Each symptom must be phrased differently each time.<br>• Example:<br>-- "Loss of appetite" → "Hasn't been eating well lately." / "Leaves food behind and seems reluctant to eat." / "Has a reduced appetite and eats much less than usual."<br>-- "Lethargy" → "Seems more sluggish than before." / "Not as active and playful as usual." / "Lies down most of the time and avoids movement."<br>• Avoid repeating identical expressions.<br>• Generate the consultation record as a single, naturally flowing consultation entry, rather than in a systematically structured format.<br>**Example:**<br>[Consultation Record 1]<br>[Consultation Record 2]<br>Selected Symptoms: [Symptoms]<br>Generated Consultation Record: |

**TABLE 2. Examples of real-world and generated consultation data.**

| | Example |
|---|---|
| **Real-World Data** | S) Does not eat real food at home, energy and appetite are good, does not eat the solids of K/D food and only eats the broth, eats chicken breast, snacks, and sweet potatoes well, had severe diarrhea the day after discharge and received subcutaneous fluids that day, had mucus-like stool early the next morning and no defecation since, no vomiting, water intake does not seem to be high. |
| **Generated Data** | The patient has recently been experiencing frequent episodes of losing consciousness and collapsing. The pet has also been observed circling and often keeping its head tilted to one side. Additionally, walking appears to be difficult, with noticeable leg weakness causing instability, suggesting potential motor dysfunction. The pet seems more irritable than usual, reacting sensitively even to minor stimuli. |

## C. DEEP-LEARNING-BASED CONSULTATION ASSISTANCE SYSTEM

Deep learning-based consultation assistance systems have been actively researched to support physicians in making accurate diagnoses and treatment plans. The study by Kim et al. [64] designed a deep learning model that improved the accuracy of lung cancer diagnosis by analyzing respiratory cytology images, serving as an auxiliary tool for pathologists. Similarly, Sharma et al. [65] conducted a comprehensive review of research on deep learning-based diagnosis and prognosis of Alzheimer's disease. With advancements in LLMs, these systems have been increasingly applied in medical consultation settings, leveraging natural language understanding and reasoning capabilities [66]. In particular, LLMs can employ chain-of-thought reasoning [67] to provide logical justifications for generated responses. This capability enhances their ability to analyze complex medical data and infer relationships between patient symptoms and diseases.

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However, the reasoning of LLMs is based on the generation of learning data. Although dynamically updating the model can improve its performance compared to the previous model, such updates will cause the knowledge mastered by the model to change, which will in turn damage the reliability and consistency of the results [68]. Even if a model's response exhibits a natural logical flow, independent verification remains necessary to ensure alignment with clinical standards [69]. Thus, explainability and real-time validation of disease prediction and recommendation processes are essential in medical consultation environments. Building upon this, our framework incorporates a structured database curated by veterinary experts, enhancing interpretability throughout the consultation process. Veterinarians can examine the system's intermediate reasoning steps, such as the extraction of symptoms, the screening of diseases based on those symptoms, and the identification of final disease candidates. These stages provide clear and transparent references for clinical judgment. By presenting the basis for decisions at each step, the framework enhances explainability, especially by clearly showing which symptoms were recognized and how the set of possible diseases is gradually narrowed throughout the diagnostic process.

## III. DATASET CONSTRUCTION

We construct a dataset to evaluate the effectiveness of AVA. Section III-A explains the rationale for dataset construction, including the purpose and necessity of building a dataset tailored for AVA evaluation. Section III-B describes the dataset construction process using the disease-symptom database, detailing how disease and symptom information is organized to create meaningful evaluation data.

### A. RATIONALE FOR DATASET CONSTRUCTION

The existing consultation record dataset within veterinary EHRs is unstructured and lacks consistency, which can create challenges in the symptom extraction process. When writing veterinary EHRs, veterinarians often use memo-style notes or abbreviations, leading to inconsistencies in how symptoms are documented across consultation records. This lack of standardization can hinder the system's ability to accurately extract symptoms, emphasizing the need for more consistent and well-organized consultation records. Accordingly, we construct a dataset to assess how accurately symptom extraction can be performed when consultation records follow a more organized and predefined format. After generating the data, we manually verified whether it had similar sentence structure and difficulty level as the real data. To avoid the model performance being artificially inflated or underestimated, we made necessary adjustments to the data. For example, if we found that the sentences were too simple or there were expressions that did not match the real clinical data, we would remove the data to manage the data quality in this way.

### B. DATASET CONSTRUCTION USING DISEASE-SYMPTOM DATABASE

In this study, we utilize a disease-symptom database certified by veterinarians, which contains 244 diseases and 464 symptoms commonly observed in veterinary consultations, as shown in Figure 2. Based on this database, we newly construct a consultation records dataset. For each disease, we generate five samples and we compose each consultation record by randomly selecting at least one and up to the maximum number of symptoms associated with the corresponding disease. In addition, we adopt the data generation approach proposed in a previous study [70] and generated consultation records using LLM (specifically, GPT-4o). During this process, we provided examples from real-world consultation records as few-shot demonstrations and rephrased the symptom names from the database into diverse expressions that may appear in actual consultation records. We present the prompt used for data generation in Table 1. We also provide examples of both real-world consultation record and the generated consultation record in Table 2.

## IV. METHODOLOGY

Our framework, AVA, leverages real-time consultation records to assist veterinarians during consultations (Figure 3). Section IV-A describes the method for extracting symptoms from consultation records. Section IV-B explains the approach for predicting diseases. Section IV-C presents the method for recommending questions during the consultation process.

### A. EXTRACTING SYMPTOMS IN CONSULTATION RECORD

Symptom extraction is a key factor in disease prediction, as symptoms and diseases are closely related. An accurate extraction process is therefore essential [71]. In this study, we design a prompt incorporating symptom extraction examples based on ICL [59], following the approach proposed in previous medical domain research [28]. The prompt consists of three components: task description, example, and input. To facilitate symptom extraction, we embed the official symptoms in a structured format within the prompt. Using this ICL-based prompt, we employ GPT-4o to extract symptoms from consultation records and identified them by comparing them with 464 official symptoms from a disease-symptom database. In addition, we design the prompt to correct expressions that were similar but did not exactly match the official symptom terms, thereby improving the effectiveness of symptom extraction. The prompt for symptom extraction is presented in Table 3.

### B. RELIABLE AND SCALABLE DISEASE PREDICTION

AVA predicts specific diseases by comparing symptoms extracted from consultation records with a disease-symptom database. After extracting symptoms, AVA matches them with the database to identify all related diseases and generates the initial disease candidates. AVA then selects the Top-$K$ diseases based on their occurrence frequency. In real-world settings, veterinarians can select the value of K based on clinical judgment, such as the complexity of the diagnosis or the

**TABLE 3.** The prompt used for symptom extraction from consultation records in AVA.

| | Prompt |
|---|---|
| **Symptom Extraction** | You are an AI model designed to analyze veterinary medical records and extract symptoms observed in animals. Below is the official list of symptoms. Extract only those symptoms from the medical records that exactly match the official symptom list.<br>**Official Symptom List:** [Symptoms]<br>**Instructions:**<br>• Extract symptoms that the animal is actually experiencing, using the exact terminology from the official symptom list.<br>• Select only symptoms that exactly match those in the official symptom list.<br>• Exclude symptoms that are mentioned but not actually observed in the animal.<br>• If the same symptom appears multiple times, list it only once.<br>• **Output Format:** List one symptom per line.<br>• **Ordering:** Maintain the order in which symptoms appear in the medical records.<br>• If no symptoms are present in the medical records, output "No symptoms."<br>• Provide the extracted list of symptoms.<br>**Example Input:** [Record 1]<br>**Example Output:** [Symptoms 1]<br>**Actual Input:** [Records]<br>**Actual Output:** |

**TABLE 4.** Examples of recommend consultation questions.

| Recommend | Example |
|---|---|
| **Frequency** | Does the patient's pet exhibit any of the following symptoms: holding the neck stiffly and avoiding movement, lowering the head while raising the hindquarters, coughing, persistent bleeding that does not stop, fever, or straining the neck excessively? The most frequently observed symptom is loss of appetite/decreased appetite. |
| **Division** | Does the patient's pet exhibit any of the following symptoms: front leg paralysis, pain when touched, swelling in the elbow area, pain when running or sitting, abnormal gait, loss of spatial orientation, coughing, fever, or excessive tension in the neck? The most optimally categorized symptom is dragging the legs. |

number of symptoms. If the number of diseases is k or fewer, AVA directly uses them as the final predicted diseases. When multiple diseases share the same occurrence frequency, AVA prioritizes diseases with a higher ratio of matched symptoms to their total symptoms. For example, consider two diseases that each have six matched symptoms. If disease A has six matched symptoms out of nine total symptoms (66.7%) and disease B has six matched symptoms out of 33 total symptoms (18.2%), AVA prioritizes disease A. Although the number of matched symptoms is the same, disease A covers a larger proportion of its defining symptoms, suggesting it is more relevant to the consultation record. By applying this step-by-step filtering process, AVA improves the accuracy of disease prediction and enhances the effectiveness of real-time consultation assistance.

### C. SYMPTOM-BASED FOLLOW-UP CONSULTATION QUESTION RECOMMENDATION

After generating the final Top-$K$ diseases, AVA retrieves the associated symptoms for these diseases from the disease-symptom database. Based on this information, AVA assists the consultation process through symptom-based consultation question recommendation. AVA provides two strategies for question recommendation, allowing veterinarians to select the most appropriate approach. First, AVA identifies the most frequently occurring symptoms among the candidate diseases and recommends consultation questions to verify their presence, helping veterinarians check critical symptoms that may not have been explicitly mentioned in the consultation record. Alternatively, AVA identifies the most decisive symptom for distinguishing between the candidate diseases and recommends a targeted question about this symptom, enabling veterinarians to efficiently rule out or confirm candidate diseases. Through these strategies, AVA assists veterinarians in formulating key consultation questions focused on core symptoms and effectively narrowing down candidate diseases, ultimately enhancing the efficiency of real-time consultations. The examples are presented in Table 4.

## V. EXPERIMENT

### A. DATASETS

We constructed the dataset following the process detailed in Section III-B. The dataset comprises 976 training samples, 244 validation samples, and 244 evaluation samples. Pre-trained language models (PLMs) were trained on the constructed training and validation datasets, and their performance was evaluated on the 244 evaluation samples by selecting the best checkpoint. In contrast, AVA and other LLM-based models were evaluated on the same 244 evaluation samples, utilizing one sample from the training data as an in-context example for symptom extraction within the prompt.

### B. EVALUATION METRICS

We evaluated the disease prediction performance of the AVA framework using Accuracy@$K$ [72], where @$K$ denotes that the Top-$K$ predictions are considered. Since AVA is designed as a veterinary consultation assistance system, its primary goal is not to predict a single correct disease but rather to provide multiple potential diseases to support veterinarians in their decision-making. Thus, Accuracy@k is an appropriate metric for assessing the practical effectiveness of AVA. In cases

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**TABLE 5.** Comparison of AVA with PLMs and other LLMs, evaluated on the validation set of the generated dataset. The best results are **boldfaced**.

| | Method | Disease Prediction | | | | | Inference Time (Sec) |
|---|---|---|---|---|---|---|---|
| | | Top-1 | Top-2 | Top-3 | Top-5 | Top-10 | |
| PLM | KM-BERT | 0.414 | 0.516 | 0.586 | 0.652 | 0.754 | **0.010** |
| | KoSimCSE-RoBERTa | 0.512 | 0.607 | 0.689 | 0.762 | 0.816 | **0.010** |
| | XLM-RoBERTa-Large | 0.004 | 0.008 | 0.012 | 0.021 | 0.041 | 0.020 |
| | KLUE-RoBERTa-Large | 0.004 | 0.008 | 0.012 | 0.021 | 0.041 | 0.027 |
| LLM | Qwen2.5-32B-Instruct | 0.172 | 0.172 | 0.176 | 0.238 | 0.357 | 9.504 |
| | EXAONE-3.5-32B-Instruct | 0.180 | 0.201 | 0.217 | 0.271 | 0.328 | 4.585 |
| Ours | AVA | **0.824** | **0.885** | **0.914** | **0.934** | **0.959** | 1.551 |

**TABLE 6.** Comparison of performance with symptom extraction and without symptom extraction. The best results are **boldfaced**. Symptom extraction accuracy is not reported for the without symptom extraction setting because symptom extraction is not performed in this case.

| | Method | Disease Prediction | | | | | Symptom Extraction |
|---|---|---|---|---|---|---|---|
| | | Top-1 | Top-2 | Top-3 | Top-5 | Top-10 | Accuracy |
| w/o Symptom Extraction | Qwen2.5-32B-Instruct | 0.172 | 0.172 | 0.176 | 0.238 | 0.357 | – |
| | EXAONE-3.5-32B-Instruct | 0.180 | 0.201 | 0.217 | 0.271 | 0.328 | – |
| | GPT-4o | 0.184 | 0.246 | 0.320 | 0.320 | 0.320 | – |
| w/ Symptom Extraction | Qwen2.5-32B-Instruct | 0.717 | 0.783 | 0.807 | 0.832 | 0.885 | 0.576 |
| | EXAONE-3.5-32B-Instruct | 0.689 | 0.779 | 0.836 | 0.865 | 0.930 | 0.648 |
| | AVA (Ours) | **0.824** | **0.885** | **0.914** | **0.934** | **0.959** | **0.799** |

where AVA retrieves fewer than $k$ diseases, the evaluation was performed based on the total number of available predictions (i.e., Top-$N$, where $N \leq k$). This adaptive evaluation approach ensures a consistent performance assessment, regardless of variations in the number of predicted diseases.

### C. EXPERIMENTAL DETAILS

We implemented the models and baselines using PyTorch [73] and HuggingFace [1]. We conducted a series of experiments with various hyperparameters to enhance the accuracy of the models. For the PLMs, we conducted experiments with batch sizes [8, 16, 32], learning rates [1e-4, 5e-5, 1e-5, 5e-6, 1e-6], and epochs [20, 30, 40]. In the case of LLMs, we performed generation and inference with the temperature [74] fixed at zero and applied greedy decoding. We conducted all experiments on two NVIDIA A6000 GPUs.

### D. BASELINES

In this study, we evaluated the performance of AVA by comparing it with both PLMs and LLMs. For the PLM baselines, we selected models optimized for the Korean language, including KM-BERT [2], KLUE-RoBERTa-large [3], KoSimCSE-RoBERTa [4], and the multilingual XLM-RoBERTa-large [5]. Notably, KM-BERT is a model pre-trained on a Korean medical corpus based on KR-BERT [6], utilizing data from medical textbooks, health information news, and medical research papers. Each PLM was fine-tuned for the disease prediction task, but no additional tuning was applied for symptom extraction. For the LLM baselines, we selected Qwen2.5-32B-Instruct [7] and EXAONE-3.5-32B-Instruct [8], both of which were pre-trained on Korean data. To accommodate resource constraints in the experimental environment, we applied 4-bit quantization [75] to these LLM baselines.

[1]https://huggingface.co/
[2]https://huggingface.co/madatnlp/km-bert
[3]https://huggingface.co/klue/roberta-large
[4]https://huggingface.co/BM-K/KoSimCSE-roberta
[5]https://huggingface.co/FacebookAI/xlm-roberta-large
[6]https://huggingface.co/snunlp/KR-BERT-char16424

### E. EXPERIMENTAL RESULTS

Table 5 presents the experimental results comparing the performance of the baseline models and AVA on the dataset. The proposed method, AVA, outperforms all baseline models in terms of accuracy, achieving 91.4%, 93.4%, and 95.9% for Top-3, Top-5, and Top-10 predictions, respectively. With an average inference time of 1.551 seconds, AVA is 1.141 seconds slower than the fastest PLM baselines, KM-BERT and KoSimCSE-RoBERTa, yet it still outperforms the LLM baselines and remains practical for real-time processing.

## VI. ANALYSIS

### A. EFFECTIVENESS OF SYMPTOM EXTRACTION

Table 6 presents the results comparing model performance with and without symptom extraction. Models without symptom extraction exhibited significantly lower performance than those incorporating symptom extraction, emphasizing its critical role in improving accuracy. Notably, AVA utilizing GPT-4o achieved the best performance among models that performed symptom extraction, attaining a symptom extraction accuracy of 79.9% and the highest scores in disease prediction. These results demonstrate that GPT-4o played a crucial role in enhancing overall performance.

[7]https://huggingface.co/Qwen/Qwen2.5-32B-Instruct
[8]https://huggingface.co/LGAI-EXAONE/EXAONE-3.5-32B-Instruct

**TABLE 7. Comparison of AVA with LLMs in real-world consultation records. The best results are boldfaced.**

| Method | Disease Prediction | | | | |
|---|---|---|---|---|---|
| | Top-1 | Top-2 | Top-3 | Top-5 | Top-10 |
| Qwen2.5-32B-Instruct | 0.168 | 0.277 | 0.337 | 0.386 | 0.475 |
| EXAONE-3.5-32B-Instruct | 0.079 | 0.168 | 0.168 | 0.347 | 0.505 |
| AVA (Ours) | **0.238** | **0.307** | **0.386** | **0.436** | **0.515** |

### B. EVALUATION OF LLM-BASED PERFORMANCE ON REAL-WORLD CONSULTATION RECORD

Table 7 presents the disease prediction performance of three LLMs applying the AVA framework to real-world consultation records. We evaluated the models using 101 consultation records, randomly sampling five records per disease to ensure a balanced representation. Overall, all models exhibited lower performance compared to their results on consultation records generated from the disease-symptom database. Nevertheless, AVA achieved the highest accuracy among the tested models, demonstrating its relative effectiveness even in real-world environments. In addition to the quantitative evaluation, we conducted a qualitative analysis to assess the reliability of symptom extraction, as the real-world consultation records did not contain labeled symptom annotations. Based on the Top-10 predictions, we categorized the consultation records into the following four groups: (1) cases where accurate symptom extraction led to successful disease prediction, (2) cases with good symptom extraction but failed disease prediction, (3) cases with inaccurate symptom extraction and (4) records lacking sufficient information for symptom extraction. Through this qualitative analysis, we identified 47 records in category (1), 29 records in category (2), 18 records in category (3), and 7 records in category (4). When we examined how the symptoms were expressed we found that the system extracted symptoms with a high probability when the consultation records used exact database terms, whereas its extraction performance declined when the same symptoms appeared in varied paraphrased forms. Also, in category (2) symptom extraction succeeded but disease prediction failed because the records mentioned too few symptoms or included additional symptoms that were not present in the database, which limited further extraction (Table 8). Given the overall strong performance of symptom extraction, these findings suggest that AVA's effectiveness could further improve when applied to consultation records containing richer and more detailed symptom information.

### C. NEGATION SENSITIVITY IN SYMPTOM EXTRACTION

We evaluated AVA's symptom extraction accuracy under linguistic ambiguity, focusing on negation. Five symptoms from the disease symptom database were selected, and 100 consultation records that negated those symptoms were created. Half of the records used the exact database terms in an explicit negative form, while the other half used varied paraphrased negatives. Each record contained five negated symptoms, yielding 250 symptom instances per condition. In the explicit negation condition AVA incorrectly extracted 23 of 250 symptoms (9.2%), whereas in the paraphrased negation condition it incorrectly extracted 86 of 250 symptoms (34.4%). These results show that AVA handles direct explicit negation relatively well but remains vulnerable to semantically similar negations expressed in more varied language.

### D. VALIDATION VIA STATISTICAL TESTS

Table 9 presents McNemar chi-square statistic and $p$-values that compare disease prediction accuracy with versus without symptom extraction under the Top-3, Top-5, and Top-10 settings. In every setting the observed chi-square exceed the conventional significance threshold by a wide margin, and all $p$-values are less than 0.05, decisively rejecting the null hypothesis of no performance difference. Notably, the AVA model shows the strongest effect in the Top-10 scenario, suggesting that symptom-based consistency is particularly effective at pruning inappropriate diseases from longer candidate lists. Consistent improvements for Qwen-2.5-32B-Instruct, EXAONE-3.5-32B-Instruct, and AVA alike confirm that the gains stem from the symptom-extraction step itself rather than model-specific biases.

## VII. DISCUSSION

In this study, we explored both the potential and limitations of deep-learning-based consultation assistance systems. AVA demonstrated strong performance in disease prediction and symptom recommendation, highlighting its effectiveness in supporting clinical decision-making. However, several challenges must be addressed for its practical application in real-world consultation settings. In particular, our findings indicate that the unstructured nature of consultation records and the quality of the data significantly impact AVA's performance. Thus, systematic data entry and management are essential to ensure stable operation in practice.

One of AVA's greatest strengths lies in its database-driven architecture. Unlike existing deep-learning-based consultation support systems that require additional training to incorporate new diseases or symptoms, AVA can immediately reflect updates simply by modifying the database. This significantly reduces maintenance costs and enables veterinarians to directly manage the database, allowing them to improve system performance in real-time, thereby enhancing its practicality. Notably, since each veterinarian may have a unique style of writing consultation records and may use different terminologies, continuously refining and updating symptom names in the database to align with each veterinarian's recording style can improve the accuracy of symptom matching and further enhance AVA's usability in clinical practice. Moreover, AVA fosters trust by transparently supporting the veterinarian's decision-making process rather than making independent judgments as a deep-learning-based model.

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**TABLE 8. Examples of consultation records for each evaluation criteria, with the number of data, representative cases, and extracted symptoms.**

| Evaluation Criteria | Number of Data | Example | Extracted Symptoms |
|---|---|---|---|
| **Accurate symptoms and correct results** | 47 | S) HPI: Due to the owner's circumstances, the patient was cared for by someone else.<br>Approximately 2 years ago, a perianal mass was detected, which grew, ruptured, and caused persistent bleeding.<br>1 year ago, pre-surgical bloodwork revealed elevated liver values, leading to surgery postponement and liver medication administration.<br>Prevention: No heartworm prevention for over a year.<br>Diet: Kibble.<br>Environment: Surgery history: None. Medications: None.<br>...<br>A) Extensive perianal mass infiltration, adrenal tumor severely invading the CdVC, renal veins, and retroperitoneal space, making surgery unfeasible, with a high risk of sudden death due to thromboembolism.<br>P). | Ulceration, bleeding, foul odor around the anus, anal bleeding |
| **Accurate symptoms and incorrect results** | 29 | FHNO surgery was performed on January 8, 2019, and 2021.<br>During the first surgery, a hip surgery was conducted, but a follow-up CT scan revealed additional issues, leading to a second surgery.<br>This time, dislocation occurred due to external trauma.<br>Postoperatively, the patient frequently exhibits dorsal paw placement.<br>The patient walks less than usual and tends to sit frequently but has no issues with urination.<br>...<br>Liver function test at this hospital: ALKP 341, ALT 171, so liver protectants were additionally prescribed.<br>Advised to monitor whether the increase is temporary post-surgery.<br>Total of 5-day medication prescribed. | Reluctance to walk, hip dislocation, difficulty sitting |
| **Inaccurate symptoms and incorrect results** | 18 | Tapering ended in early December.<br>Recheck on January 2 showed no abnormalities, and a polyvalent vaccine was administered.<br>...<br>#1 (post-meal) PDS 1.5mg/kg PO BID (4.5T), Famotidine 0.5mg/kg PO BID, Doxycycline 5mg/kg PO BID, UDCA 10mg/kg PO BID, Clopidogrel 1mg/kg PO BID. | Decreased energy, anemia, sticky urine, weight loss |
| **No information** | 7 | [Hospitalization Monitoring] – Appetite test with dry food scheduled from today.<br>No vomiting or diarrhea.<br>Pain response improved, so butor reduced from QID to BID IV.<br>O) fPL decreased: 92.32 $\rightarrow$ 26.09 $\rightarrow$ 14.73.<br>Electrolytes/gases, BG, ALT, Phos all WNL. | - |

**TABLE 9. Comparison of disease-prediction performance with and without symptom extraction, reported as McNemar chi-square statistics and p-values.**

| Method | Top-3 | | Top-5 | | Top-10 | |
|---|---|---|---|---|---|---|
| | chi-square | $p$-value | chi-square | $p$-value | chi-square | $p$-value |
| Qwen2.5-32B-Instruct | 142.86747 | 6e-33 | 133.91720 | 6e-31 | 113.20408 | 2e-26 |
| EXAONE-3.5-32B-Instruct | 139.88344 | 3e-32 | 132.23270 | 1e-30 | 139.41290 | 4e-32 |
| AVA (Ours) | 141.10738 | 2e-32 | 148.02632 | 5e-34 | 154.02532 | 2e-35 |

## VIII. CONCLUSIONS

In this study, we propose AVA, a framework that leverages GPT-4o and a disease-symptom database to assist veterinarians through disease prediction and recommend consultation questions. The AVA database-driven approach allows real-time updates without retraining, and veterinarians can directly modify the database, ensuring flexibility and scalability. These advantages make AVA a practical and adaptable consultation assistance tool for veterinary medicine.

## LIMITATION

AVA has demonstrated its ability to assist in veterinary consultations, predicting diseases and recommending relevant consultation questions, but there are still several limitations. 1) In terms of disease symptom databases, first of all, the framework relies on a pre-built disease-symptom database, and its performance is largely affected by the quality and coverage of the database. If certain diseases or symptoms are missing from the database, it may lead to inaccurate identification or prediction, so the database needs to be continuously updated and maintained. 2) In terms of consultation data synthesis, first of all, although the consultation records used in this study were generated based on expert-reviewed databases and real data, they were not directly reviewed by veterinarians, and only manually verified whether their sentence structure and difficulty were close to real scenarios. These records may not fully cover the co-occurrence patterns and special cases of symptoms. Secondly, from a non-expert perspective, it is difficult to include clinically important details—such as symptom intensity and onset timing—when generating synthetic data. In addition, although different LLMs were used for dataset-level experiments, concerns about fairness still remain, as GPT was used both to generate the synthetic data and to extract symptoms from it. 3) In terms of the AVA system, first of all, the current system does not have a clear mechanism to deal with symptom extraction failure or uncertainty. In the future, it is planned to introduce a user intervention mechanism to enhance the system's review and modification capabilities. Secondly, the feasibility and effectiveness of the two strategies of "most frequent" and "most decisive" proposed in the consultation question recommendation still need to be further verified by veterinarians. Finally, as an autonomous diagnostic system, the current level of symptom extraction accuracy may not yet meet the demands of real clinical practice. In particular, negations of symptoms found in consultation records—especially those that rephrased symptom terms—can sometimes lead to the incorrect extraction of unrelated symptoms. However, AVA is intended to function as a decision support framework, aiming to provide auxiliary information for veterinarians to use as a reference. 4) In terms of experiments, first of all, although there is no direct duplication between the evaluation samples and the ICL demonstration samples in the experiment, the similarity in expression may still bring the risk of data leakage. Secondly, although the proposed method does not depend on a specific language, all experiments in this study were conducted in a Korean environment and have not been directly verified in other language environments. Finally, the evaluation results on real consultation records are generally relatively poor. The reason is that the actual consultation records lack symptom information and contain a large amount of irrelevant information, which also poses a challenge to the model's symptom extraction and reasoning capabilities, affecting its performance in real cases.

## ETHICAL CONSIDERATION

Using AVA as a consultation assistance framework may raise ethical concerns, particularly if system malfunctions or mispredictions negatively affect the health of companion animals. In addition, legal and professional regulations must be taken into account. According to the Korean Veterinary License Act, all diagnostic and prescription activities must be directly performed by a licensed veterinarian. This implies that even if pet owners attempt to use AVA for preliminary consultation or self-assessment, such use could result in legal and ethical issues. Therefore, AVA is designed exclusively as a support tool for licensed veterinarians. The primary goal of this study is not to replace human expertise, but to alleviate the workload of veterinarians. Final decision-making authority remains with human professionals, and AVA functions not as an autonomous consultation system, but as a supportive tool to assist in clinical decision-making. These limitations of AVA usage are closely connected to broader ethical considerations surrounding the application of LLM technologies in the medical field. In practice, the use of LLMs in healthcare has raised ongoing concerns about privacy, bias, and patient consent [76]. Furthermore, when dealing with human medical data, the process of identifying and removing sensitive information from EHRs has been emphasized as a critical issue [77]. Likewise, veterinary clinical data should also be managed with a comparable level of care and caution.

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**YOUN-GYU JIN** received the B.S. degree in the school of computer science and information engineering from The Catholic University of Korea, Bucheon, South Korea, in 2024, where he is currently pursuing the M.S. degree in Artificial Intelligence. His research interests include natural language processing, deep learning, parameter-efficient fine-tuning, and conversational artificial intelligence.

**GUIXIN WU** received the B.S. degree in Automation in the department of International Education from Henan University of Science and Technology, Henan, China, in 2022. He is currently pursuing his M.S. degree in Artificial Intelligence from The Catholic University of Korea, Bucheon, South Korea. His research interests include natural language processing, computer vision.

**JU-WON SEO** received the B.S. degree in Energy & Environmental Engineering from The Catholic University of Korea, Bucheon, South Korea, in 2024, where he is currently pursuing the M.S. degree in Artificial Intelligence. His research interests include natural language processing, large language models, large multimodal models, and deep learning for pet care.

**SEONG-JIN PARK** received the B.S. degree in the department of Mathematics and the school of Computer Science and Information Engineering from The Catholic University of Korea, Bucheon, South Korea, in 2025, where he is currently pursuing the M.S. degree in Artificial Intelligence. His research interests include large language models, vision-language models, and autonomous agents.

**SUNG-HO HUR** majored in Information and Computer Engineering at Ajou University in Suwon in 2001. He founded IntoCNS in 2007 and has grown it into the No. 1 company in the domestic animal-related system market share for 18 years, leading the veterinary IT industry in Korea. Based on the know-how accumulated over the years, he created the companion animal hospital management system IntoVet GE (EMR), and based on this, he built a 'smart animal hospital system' and developed and linked various additional systems. Currently, he is continuously expanding the pipeline to secure various data on companion animals at home and abroad.

**DINARA ALIYEVA** received a B.S. degree in Computer Science from Korea Advanced Institute of Science and Technology (KAIST), in 2015, and M.Eng. degree in Computer Science and Engineering from Korea University, in 2018. She is currently pursuing her M.S. degree in Computer Science at the University of North Carolina, Chapel Hill. Her research interests include Multimodal Learning, Explainable AI and Reinforcement Learning.

**JUN-HYUNG PARK** received B.S. and Ph.D. degrees in Computer Science and Engineering from Korea University, Seoul, South Korea, in 2018 and 2023, respectively. Since 2024, he has been an assistant professor in the Division of Language & AI at Hankuk University of Foreign Studies, Seoul, South Korea. His primary research lies in natural language processing and efficient artificial intelligence, including efficient training and augmentation methods on language models and visual language models.

**KANG-MIN KIM** received the B.S. degree in Computer Engineering from Kyung Hee University, Yongin, South Korea, in 2016, and a Ph.D. degree in Computer Science and Engineering from Korea University, Seoul, South Korea, in 2021. Since 2021, he has been an Assistant Professor with the Department of Data Science and the Department of Artificial Intelligence, The Catholic University of Korea, Bucheon, South Korea. His research interests include natural language processing, large-scale text classification, commonsense reasoning, self-supervised learning, weakly supervised learning, and intelligent systems.

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